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Module 3
Emerging Technologies

Biomass Resources

Biomass is the general term for living material. The energy obtained from biomass is known as biomass energy. Animals feed on plants and plants grow through photosynthesis process using solar energy. Thus photosynthesis is primarily responsible for biomass energy. To use biomass energy, the initial biomass may be transformed by chemical/biological process to produce gaseous &liquid fuels. On combustion it reacts with oxygen to release heat.

Example: wood, charcoal, producer gas, sugarcane, sugar beet, cassava, sunflower, palm, castor, soybean, Rice husk, straw, coconut shell, seaweed, algae, Municipal solid waste, sewage etc.

Advantages of Bio-mass Energy

- Renewable
- Storage of energy in usable form
- Emission of pollutants is less than fossil fuels
- Commercial use of biomass reduces waste disposal problem
- Use of biogas plants, leads to improved sanitation, better hygienic conditions in rural areas
- Can extend capacity as required.

Disadvantages of Bio-mass Energy

- Low energy density
- Labour intensive process
- Not feasible for all locations
- Availability of biomass
- Not suitable for varying and peak load

Principle of Biomass Energy Generation

Biomass energy is generated by converting organic materials such as wood, agricultural waste, animal manure, and biodegradable waste into useful forms of energy. The basic principle relies on the fact that biomass contains stored chemical energy captured from sunlight through photosynthesis. When this biomass is burned or processed, the chemical energy is released as heat, which can be used directly or converted into electricity. Biomass can be converted through processes like combustion, gasification, pyrolysis, anaerobic digestion, and fermentation. These processes produce heat, biogas, or biofuels, which can then run boilers, turbines, engines, or generators to produce electricity. Thus, biomass energy generation works on the principle of transforming stored solar energy in organic matter into usable thermal, mechanical, or electrical energy.

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Bio fuels

Biomass is organic material that reacts with oxygen in combustion and natural metabolic process to release heat. Sometimes, it is used as such in its original form but more often it is transformed into modern energy forms such as liquid and gaseous fuels, electricity, and process heat to provide energy services needed by rural and urban populations and also by industry. Some of its forms available to users are discussed below:

- ***Bio ethanol [C_2H_5OH]***

It is a colourless liquid bio fuel. It is derived from wet biomass containing sugars (sugarcane, sugar beet etc), starch (grains, and tubers like potato) or cellulose.Ethanol is largely produced from sugar cane. 22% anhydrous ethanol + gasoline is used effectively in vehicles. It is being used by the large number of automobiles in the world.

- ***Bio gas***

Biogas is produced from organic waste of plants and animals in a biogas fermentor or digester. N2 rich sludge is also produced as a by-product which is used as fertilizer.The biogas contains 60% CH_4, 30-40% CO_2, 5-10% H_2, 0.5-0.7% N_2 and traces of $O_2$and H_2S.It is used for cooking, lighting, heating and operating small IC engines. It is unlikely to be used for mobile vehicles on large scale because of low pressure and high inert fraction.

- ***Producer Gas***

Woody matter such as crop residue, wood chips, bagasse (fibrous residue of sugar cane after juice extraction), rice husk, coconut shell, etc., can be transformed to producer gas (also known as synthesis gas, syn gas, wood gas, and water gas or blue gas) by a method known as thermal gasification of solid fuel. The composition of gas produced depends upon the type of biomass and the design of gasifier. For wood chips as input the typical gas composition is 19 per cent CO, 18 per cent H2, 1 per cent CH4, 11 per cent CO2 and the rest N2. This can be used to fuel IC engines (diesel, dual fuel mode engines) for irrigation pumps, motor vehicles, and small-scale power generation or to produce process heat.

- ***Bio-diesel***

Some vegetable oils, edible as well as non-edible can be used (after some chemical processing) in pure form or its blend with petroleum diesel as fuel in diesel engine. Bio-diesel is simple to use, biodegradable, nontoxic, and essentially free of sulphur and aromatics.

The raw vegetable oil is upgraded as bio-diesel through a chemical process called trans-esterification whereby the glycerine is separated from the animal fat or vegetable oil. The process leaves behind two products—methyl (or ethyl) esters (the chemical name for bio-diesel) and glycerine (a valuable by-product usually sold to be used in soaps and other products). Bio-diesel can be produced from vegetable oils, animal fats or recycled restaurant greases.Jatropha and Karanj are the most promising bio-diesel resources.

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- ***Fuel Pellets and Briquettes***

Crop residues such as straw, rice husk etc. and waste wood are pressed to form lumps, known as fuel pellets or briquettes and used as solid fuel. The purpose is to reduce moisture content and increase the energy density of biomass making it more feasible for long distance transportation.

- ***Fuel Wood (Virgin Wood)***

Wood is the most obvious and oldest source of biomass energy. This was the main source of energy used by mankind for centuries. Direct combustion is the simplest way to obtain heat energy. Its energy density is 16–20 MJ/kg. It can also be converted to more useful forms such as charcoal or producer gas. About half of the world population depends on fuel wood or other biomass such as cow dung, crop residue etc. for cooking and other domestic uses. Because of inefficient use of fuel wood in conventional household stove, only 5 per cent heat is utilized. The remainder is lost due to wind, incomplete combustion, radiation losses and other losses resulting from mismatch of fire and pot size. Considerable energy is also wasted in evaporation from uncovered pot and from use of wet fuel. Smoke, which is in fact unburnt tar and carbon, is a health hazard. There is little control overthe rate at which wood is burnt.

- ***Charcoal***

Charcoal is a clean, dry, solid fuel of black colour. It has 75–80 per cent carbon content and has energy density of about 30 MJ/kg. It is obtained by carbonization process of woody biomass to achieve higher energy density per unit mass, thus making it more economical to transport. It can be used as fuel in domestic environment as it burns without smoke. In the industrial sector it is used in specialized applications where specific fuel characteristics are required, such as high carbon and low sulphur content. Chemical grade charcoal has many uses in laboratory and industrial chemical processes. It is also used for making high quality steel.

Biomass Conversion Technologies

There are many different ways of extracting energy from biomass. These energy conversion technologies may be grouped into four basic types:

- Physical method,
- Incineration (direct combustion),
- Thermo-chemical method
- Biochemical method.

Physical Method

The simplest form of physical conversion of biomass is through compression of combustible material. Its density is increased by reducing the volume using compression through the processes called ‘briquetting’ and ‘pelletization’. Fuel oils can be extracted from

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plant products by expelling them. Also light hydrocarbons may be obtained from certain plants in the same way as production of rubber.

(a) Pelletizaton

Pelletization is a process in which waste wood is pulverized, dried and forced under pressure through an extrusion device. The extracted mass is in the form of pellets, facilitating its use in steam power plants and gasification system. Pelletization reduces the moisture to about 7 to 10 percent and increases the heat value of the biomass.

(b) Briquettng

Biomass briquettes are made from woody matter (e.g. agricultural waste, sawdust, etc.). They serve as replacement for fossil fuels such as oil or coal, and can be used to heat boilers in manufacturing plants. Burning a wood briquette is far more efficient than burning firewood. Moisture content of a briquette can be as low as 4 per cent, whereas in green firewood it may be as high as 65 per cent.

Briquetting is brought about by compression and squeezing out moisture and breaking down the elasticity of the wood and bark. If elasticity is not sufficiently removed, the compressed wood will regain its pre-compression volume. Densification is carried out by compression under a die at high temperature and pressure. It is a process similar to forming a wood pellet but on a larger scale. There are no binders involved in this process. The natural lignin in the wood binds the particles of woodtogether to form a solid piece.

Sawdust briquettes have developed over time with two distinct types: those with holes through the centre, and those which are solid. A solid briquette is formed using a piston press, which sandwiches the layers of sawdust together. The one with a hole is produced using a screw press. The hole is formed due to screw thread passing through the centre. The screw press briquettes are more homogeneous, have better crushing strength and better storage properties with extraordinary combustion properties due to large surface area per unit weight.

(c) Expelling Agro Products

Concentrated vegetable oils may be obtained from certain agro products and may be used as fuel in diesel engines. However, difficulties arise with direct use of plant oil due to high viscosity and combustion deposits. Therefore, these oils are upgraded by a chemical method known as trans-esterification to overcome these difficulties.

Categories of certain materials with examples are as follows:

- Seeds: Sunflower, rapeseed, soya beans, etc.
- Nuts: oil palm, coconut copra, jojoba nuts, etc.
- Fruits: olive, etc.
- Leaves: eucalyptus, etc.

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(d) Fuel Extraction

Occasionally, milky latex is obtained from freshly cut plants. The material is called exudates and is obtained by cutting (tapping) the stems or trunks of the living plants (a technique similar to that, used in rubber production).

Some plants are not amenable to tapping and, in such cases, the whole plant (usually a shrub) is crushed to obtain the product. For example, Euphorbia lathyris plant is crushed to extract hydrocarbons of less molecular weight than rubber. The extract may be used as petroleum substitute.

Incineration

Incineration means direct combustion of biomass for immediate useful heat. The heat (usually in the form of steam) produced are either used to generate electricity or provide the heat for industrial process, space heating, cooking or district heating. Furnaces and boilers have been developed for large scale burning of various types of biomass such as wood, waste wood, black liquid from pulp industry, food industry waste, and MSW. The moisture content in the biomass and wide range ofcomposition tends to decrease the efficiency of conversion. However, the economic advantage of cogeneration makes it attractive for adoption.

Thermo-Chemical

The basic thermo-chemical process to convert biomass into more valuable and/or convenient product is known as pyrolysis. Biomass is heated either in absence of oxygen or by partial combustion of some of the biomass in restricted air or oxygen supply. Pyrolysis can process all forms of organic materials including rubber and plastics, which cannot be handled by other methods.

The products are three types of fuels, usually: a gas mixture (H_2, CO, CO_2, CH_4 and N_2), oil like liquid (water soluble phase including acetic acid, acetone, methanol and non-aqueous phase including oil and tar) and a nearly pure carbon char. The distribution of these products depends upon the type of feedstock, the temperature and pressure during the process and its duration and the heating rate.

High temperature pyrolysis (~1000 °C) maximizes the gaseous product. The process is known as gasification. Low temperature pyrolysis (up to 600 °C) maximizes the char output. The process has been used for centuries for production of charcoal. The process is known as carbonization.

Liquid product is obtainedthrough catalytic liquefaction process. Liquefaction is a relatively low temperature (250–450 °C), high pressure (270 atm) thermo-chemical conversion of wet biomass, usually with high hydrogen partial pressure and also a catalyst to enhance the rate of reaction and/orto improve the selectivity of the process.

Biochemical

The process makes use of metabolic action of microbial organism on biomass to produce liquid and gaseous fuel. Two major biochemical processes are explained below:

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(a) Ethanol Fermentation (Alcoholic Fermentation):

It is the decomposition in absence of air of simple hexose sugars (sugars containing six carbon atoms per molecule, i.e. $C_6H_{12}O_6$) in aqueous solution by action of enzyme (a natural catalyst) present in yeast, in acidic conditions (pH value 4 to 5). The products are ethanol and carbon dioxide.

$$C_6H_{12}O_6 \xrightarrow[\text{Ferementation}]{32\ ^\circ C} 2C_2H_5OH + 2CO_2$$

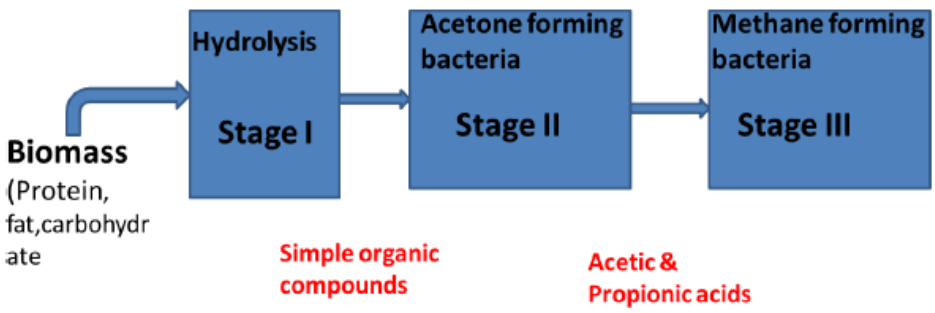
(b) Anaerobic Fermentation (Anaerobic Digestion):

This process converts decaying wet biomass and animal wastes into biogas through decomposition process by the action of anaerobic bacteria (bacteria that live and grow in absence of oxygen). Carbon present in biomass may be ultimately divided between fully oxidized CO_2 and fully reduced CH_4.

$$\text{Decaying wet biomass} \xrightarrow[\text{Anaerobic Fermentation}]{20-55\ ^\circ C} \text{Biogas (largely } CH_4 \text{ \& } CO_2\text{)}$$

The biomass material in the form of water slurry is digested by the bacteria anaerobically for several days in an airtight container. The reactions are slightly exothermic and small amount of heat (equivalent to 1.5 MJ per kg dry digestible material) is also generated that helps in maintaining favourable temperature. The process may be expedited at somewhat higher temperature. The most useful biomass materials appear to be animal manure, algae, kelp, hyacinth, plant residues and other organic waste materials with high moisture content.

Anaerobic Digestion of Biomass into Biogas (Bio-Digester)



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Stage 1:

The original organic matter containing complex compounds. (Eg: Carbohydrates, protein, fats)are broken by the influence of water known as hydrolysis to simple water-soluble compounds. The process takes about a day at 25^0C in an active digester.

Stage 2:

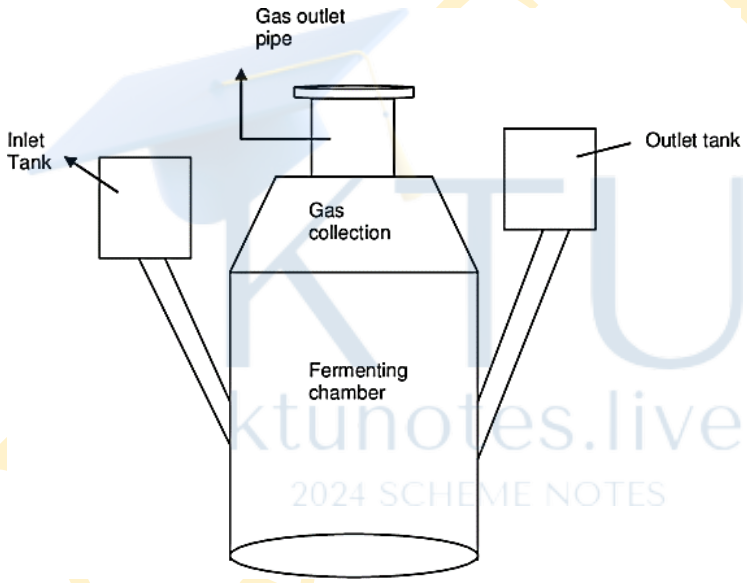
The micro-organisms of anaerobic and facultative (that can live and grow without or with oxygen) groups, together known as acid formers, produce mainly acetic and propionic acids.

Stage 3:

Anaerobic bacteria, also known as methane formers slowly digest the products available from the second stage to produce methane, carbon dioxide, a small amount of hydrogen and other gases. This process takes about two weeks at 25^0C to complete.

Digester:

The main sections of a digester are inlet tank, fermenting chamber, gas collection chamber, gas outlet pipe and outlet tank. Inlet tank is used to input the slurry into the digester. Biomass is mixed with water here. All the steps of anaerobic digestion process happen inside the fermenting chamber.



The waste sludge is carried out through the outlet tank. The formed bio-gas is stored in the gas collection chamber and take it out through gas outlet pipe.

Biomass Gasification

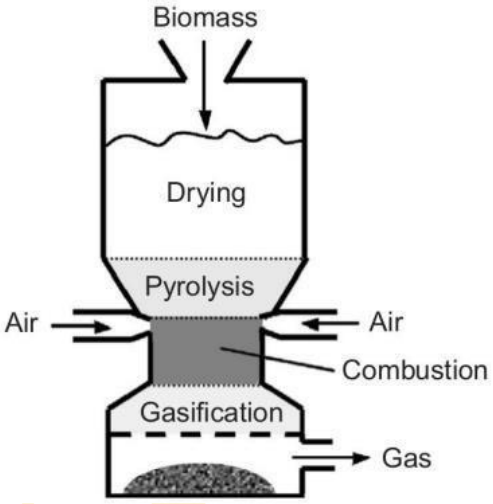
Gasifier: convert biomass into gaseous fuel by thermochemical method without any leaving solid carbon residue.

Downdraft or Co-Current Gasifiers:

In this type of gasifier, the fuel is introduced at the top of the gasifier and air is introduced at or above the oxidation zone of the gasifier. The producer gas is removed at the bottom of the apparatus, so that fuel and gas move in the same direction.

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On their way down the acid and tarry distillation products from the fuel must pass through a glowing bed of charcoal and therefore are converted into permanent gases hydrogen, carbon dioxide, carbon monoxide and methane.



Drying is the process to remove vapor content from the biomass. Pyrolysis process is the thermal decomposition of materials at elevated temperatures in an inert atmosphere. It involves a change of chemical composition.

The oxidation process occurs first, oxidation is a process which involves the addition of oxygen by adding air. Gasification is a process that converts biomass- or fossil fuel-based carbonaceous materials into gases, including as the largest fractions: nitrogen (N_2), carbon monoxide (CO), hydrogen (H_2), and carbon dioxide (CO_2).

Depending on the temperature of the hot zone and the residence time of the tarry vapors, a more or less complete breakdown of the tars is achieved.

Urban Waste to Energy Conversion

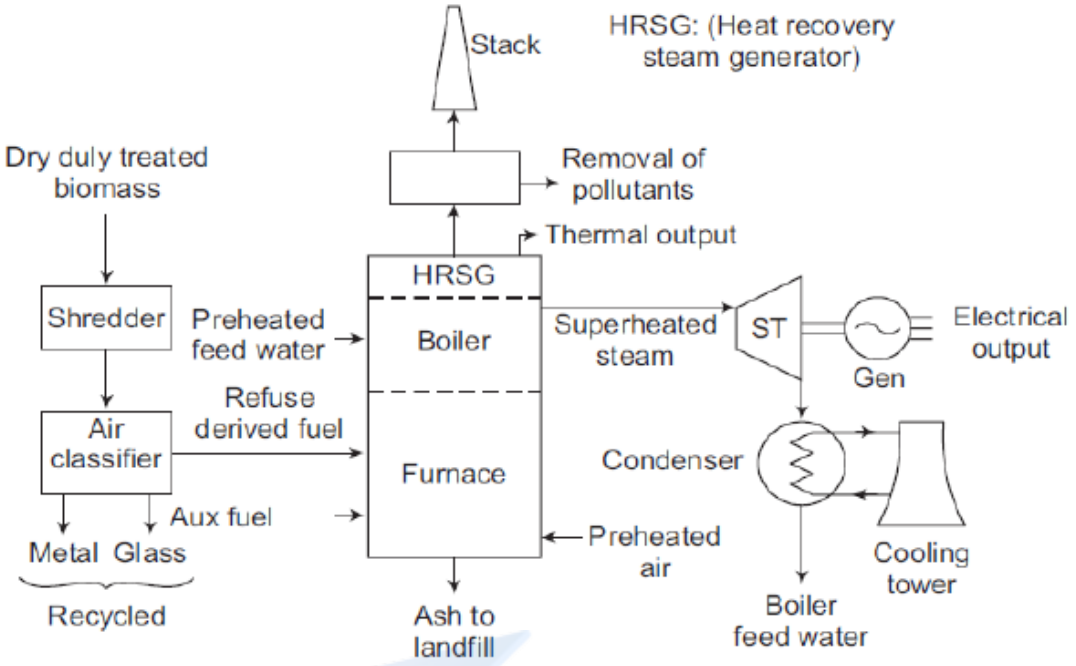
- ***MSW Incineration Plant***

Municipal solid waste is the solid waste generated by households, commercial and institutional operations and some industries. Disposal of MSW is a major problem in big cities where large quantities of waste is to be disposed of, far away from the city centres. The emerging solution is to use this waste biomass as an energy resource in a waste-to-energy conversion plant near city centre. The energy thus generated is used within city itself and only a relatively small residue of used biomass (ash etc.) is disposed away in landfills. Through incineration or gasification, electrical energy may be generated along with thermal energy for process heat.

The composition of urban waste depends on the style and standard of living of the people. The energy content of domestic refuse differs significantly from area to area. Generally, it lies in range 4.2 to 14 MJ/kg.

A block diagram of MSW-to-energy incineration plant showing the sequence of various steps is shown in Fig. The dry biomass is shredded to pieces of about 2.5 cm diameter. An air stream segregates refuse derived fuel (RDF), which is lighter from heavier metal and glass pieces. The heavier part is reclaimed and recycled. The RDF thus obtained is burnt in the furnace at about 1000 °C to produce steam in the boiler. Combustion process may be assisted by required amount of auxiliary fuel when RDF does not burn properly by itself.

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The superheated steam obtained from boiler is used in a steam turbine coupled with an alternator to produce electrical output in the same way as in a conventional thermal plant. The flue gases are discharged to atmosphere through stack after removal of pollutants such as particulate matter, SOx and NOx, etc. Heat recovery steam generator extracts maximum possible heat from flue gases to form thermal output. The ash is removed and disposed of to landfills.

Strong concerns exist about the pollutants emitted during combustion. Because of various materials contained in the waste stream, pollutants in the flue gases include derivatives of sulphur, chlorine, fluorine, nitrogen, chlorinated hydrocarbons and heavy metals.

One particular important group of pollutants, polychlorinated dibenzo-p dioxins (PCDD) and the dibenzofurans (PCDF) are possibly among the most acutely carcinogenic compounds known so far. It has been proven to cause cancer in animals and is linked with a variety of other health problems, including genetic defects. PCDD/PCDF are formed either in the combustion chamber or after combustion when the gases cool in the exhaust stack.

Disagreement continues over establishing safe levels of dioxins emission from any incinerator. It has been discovered at various levels at every incinerator tested so far. The residual ash, which has high metal concentration, is also of concern. The ash is usually disposed of in landfills; the leachate (garbage juice; liquid that results when water percolates through the landfill) must be well contained to prevent groundwater contamination.

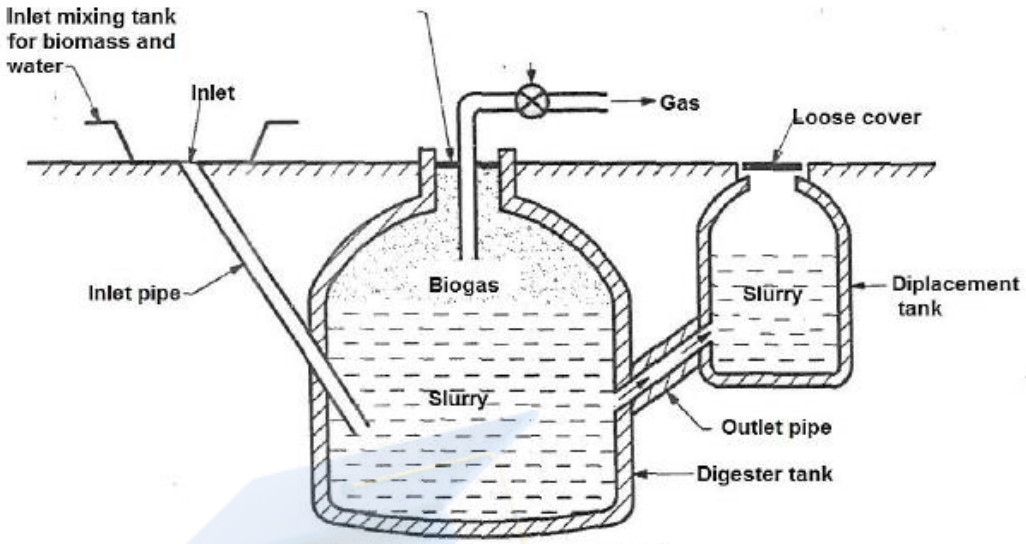
- ***Sewage to Energy Conversion***

Sewage produced by human settlements also has some energy potential. It can also be anaerobically digested (after some processing) to produce methane in the same way as with

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animal manure. The gas produced can be used to heat the digester. Though the gas production is not much and may not be sufficient for heating, but it can reduce the conventional fuel requirement.

Janta Model (Fixed Drum Type) Biogas Powerplant



Fixed dome type biogas plant

This was first developed by the planning, research and action division, Lucknow in 1978.it is an improved version of the Chinese fixed dome biogas plant.

The foundation of Janta biogas plant is laid at the base of the underground pit on a levelled ground which bear the load of the slurry as well as digester walls.The gas pressure inside the dome varies depending upon the rate of gas production and its consumption. However, the total volume of gas inside the dome remains constant due to which these are also called as constant volume type biogas plants.

The mixture of biomass and water is supplied as feed to the digester through the inlet pipe. The mixture is stirred with a handle to form slurry. The biogas generated in the digestion tank by anaerobic digestion is collected in the upper part of the digester. A fixed-dome plant comprises of a closed, dome-shaped digester with an immovable, rigid gas-holder and a displacement pit, also named 'compensation tank' or displacement tank. The gas is stored in the upper part of the digester. When gas production commences, the slurry is displaced into the compensating tank. Gas pressure increases with the volume of gas stored, i.e. with the height difference between the two slurry levels. If there is little gas in the gas-holder, the gas pressure is low.

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Advantages of Fixed Dome Type Plant

- Cost of plant is less compared to floating drum type plant.
- Loss of heat is negligible since these are constructed underground.
- No corrosion problems as in fixed drum type.
- It is maintenance free.

Disadvantages of Fixed Dome Type Plants

- Needs skilled labour to operate.
- Gas production/m^3 of digester volume is less.
- Gas is produced at variable pressure.

Site Selection Criteria of Bio Gas Plant.

- Should have minimum distance between the points of gas production and gas consumption.
- Should be constructed in open space to utilize sun energy since the biogas production is high above 25° C temperature.
- It should be constructed at least 10 m away from wells to avoid seepage of fermented slurry.
- Sufficient quantity of water along with biomass is available at site for formation of biomass slurry.
- Sufficient space is available for its operation and maintenance.

Biomass Resources Available from Nature

- ***Forests:*** Source of fuel wood, charcoal, producer gas, eucalyptus, poplar, pine.
- ***Energy crops***: Produce raw material for biofuel(ethanol) eg: sugarcane, sugar beet, cassava, grains
- ***Oil producing plants*** eg: sunflower, palm, castor, soyabean, groundnut.
- ***Agricultural residue***: Ricehusk, straw, coconut shell-used for solid fuels.
- ***Aquatic plants***: Raw material for ethanol/ biogas eg: water hyacinth, seaweed, algae
- ***Urban waste***: Municipal solid waste, sewage, Energy from MSW can be obtained from direct combustion or incineration.

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Tidal Power

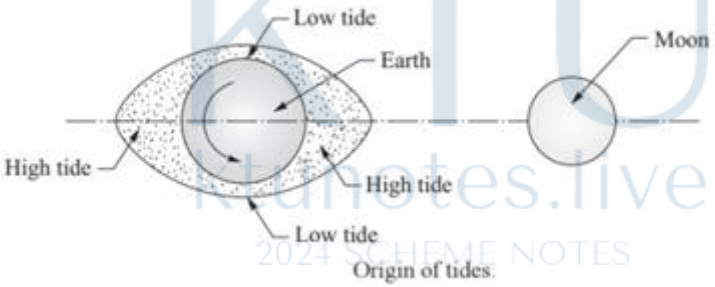
Tidal energy

Tides are produced by the gravitational attraction of the moon and the sun acting on the rotating earth. Tides are the periodic rise and fall of the water level in the oceans due to various positions of the rotating moon and sun.

Different positions of the sun and moon create two different types of tides: spring tides and neap tides. When the sun and the moon pull together whether both pulling on the same side of the earth or each on opposite sides, the result is the very high spring tides. When the sun and the moon are at 90^0 to each other the result is the lower neap tides. The period between neap and spring tide is approximately 14 days.

Moon takes 24h and 50 minutes to complete a rotation on its axis (one tidal Day). High tides / Low tides is formed about two times at a place in its one rotation. The difference in the level between the high and low tide is the Tidal range. Tidal range of 5m to 15m are considered as suitable for power generation.

As the moon is nearer to the earth, the gravitational pull of the moon is about 2.33 times stronger than that of the sun. Owing to the gravitational pull of the moon, surface water on the earth facing moon is pulled and raised towards the moon, and so water moves from poles to equator where the moon pull is maximum.

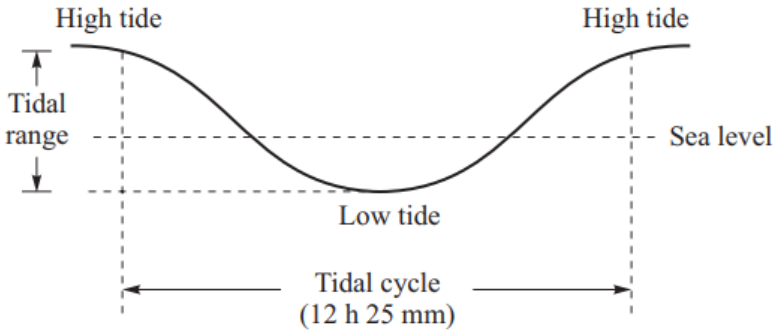


Origin of tides.

This movement of water results in the formation of a high tide at the earth's surface facing the moon. A corresponding high tide is also formed on the opposite side of the earth as solid earth is pulled away from the water surface, which results in the apparent rise of water as shown in Figure.

As moon takes 24 h and 50 min to complete a rotation on its axis, high tide is formed about two times at a place in its one rotation; that is, a high tide due to moon is formed after a period of 12 h and 25 min. The effect of gravitational pull by the sun is the same, but it is 2.2 times lesser than what is exerted by the moon.

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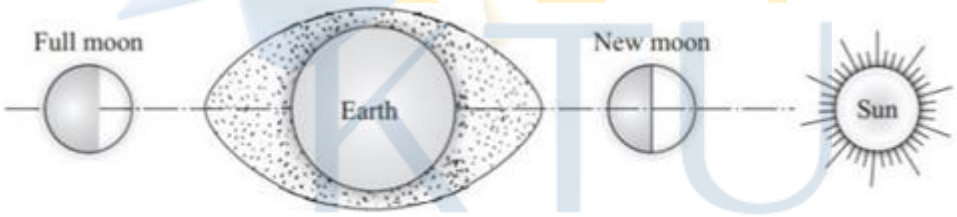


Tidal range and tidal cycle.

Earth revolves around the sun in 24 h. Owing to this slight difference in periods of rotation, solar tides move in and out of phase with lunar tides.

Spring Tides

When earth, sun and moon are aligned during rotation in a line as shown in Figure, these celestial bodies produce tides having maximum tidal heights as both solar and lunar tides are in phase. The spring tides are produced twice in a lunar month.

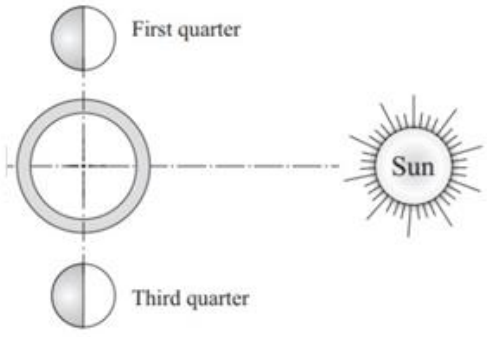


Formation of spring tides.

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Neap Tides

When sun-earth and moon-earth directions are perpendicular as shown in Figure, lunar and solar tides are completely out of phase. The celestial bodies produce tides having minimum tidal heights, which are called neap tides. Neap tides are produced twice in a lunar month.

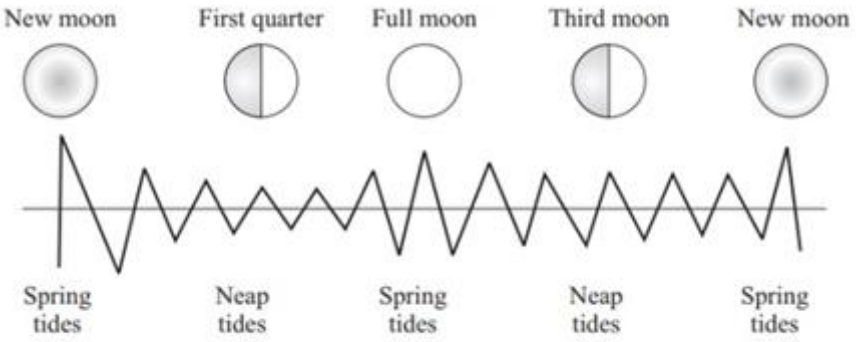


Formation of neap tides.

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Tidal Variation in A Lunar Month

The tidal variation resulting in spring and neap tides in a lunar month is shown in fig.



Tidal variation in a lunar month.

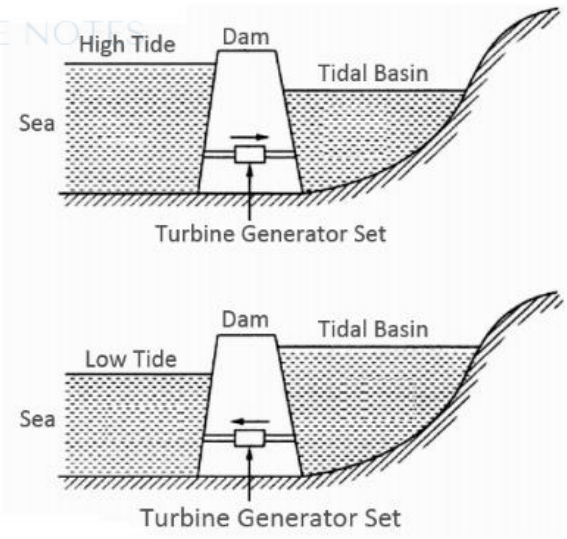
Tidal Energy

The source of tidal energy is the difference in water levels between higher flood tides and low or ebb tides. During one tidal cycle of 12 h and 25 min, the water level rises to maximum during high tides, then it starts falling to a minimum level at low tides and finally it starts rising again to a maximum level at high tides.

The difference in water levels between high tides and low tides is the potential energy of water which can be harnessed. This harnessed energy is called the tidal energy.

Basic principle of tidal power plant

A dam is constructed in such a way that a basin gets separated from the sea and a difference in the water level is obtained between the basin and sea. The constructed basin is filled during high tide and emptied during low tide passing through sluices and turbine respectively. The Potential energy of the water stored in the basin is used to drive the turbine which in turn generates electricity as it is directly coupled to an alternator.



Components of Tidal Power Plants

- Dam or Barrage.
- Sluice-ways.
- Power house.

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Barrage

The function of a dam is to form a barrier between the sea and the basin or between one basin and the other in case of multiple basins. Tidal power barrages have to resist waves whose shock can be severe and where pressure changes sides continuously. The barrage needs to provide channels for the turbines in reinforced concrete.

The location of the barrage is important, because the energy available is related to the size of the trapped basin and to the square of the tidal range. Tidal barrages require sites where there is a sufficiently high tidal range to give a good head of water – the minimum useful range is around three metres.

Sluice Ways

The sluice ways are used either to fill the basin during the high tide or empty the basin during the low tide, as per operational requirement

These may also be provided to empty the basin during low tides. Sluice ways are controlled through gates.

Power house

The turbines, electric generators and other auxiliary equipments are the main components of a power house.

The water with high potential energy is made to run through turbines to run generators for power production.

Modes of Operation of Tidal Power Plant

Ebb generation.

The reservoir or basin is filled up during high tides. The stored water is made to flow out of the reservoir to the ocean through turbines during low tides. The power generation is possible during ebb tides.

Flood generation.

The sea water is made to flow through the turbines into the reservoir or basin during high tides. This process is just the reverse of ebb generation. Power generation is possible during high tides.

Two-way generation.

The water is made to flow through turbines for power generation both during high tides from sea to basin and during low tides from basin to sea. The power generation is possible during both low and high tides.

Pumping and turbining.

The reservoir is filled using

- rise of sea water during high tides and
- pumps to pump sea water into the reservoir.

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The turbine acts as a pump-to-pump water into a reservoir during high tides and as a prime mover to the generator during low tides.

Feasibility of TPP

Tidal range: Minimum of 5m

Cost: minimum cost to construct a dam to create a basin of a required storage capacity.

Shipping Traffic: Should not cause any interruption to the shipping traffic through the estuary.

Safety: Site should be well protected, especially high wave action during storms.

Silt: The silt content in water in estuary should me small to avoid silt deposition.

Merits

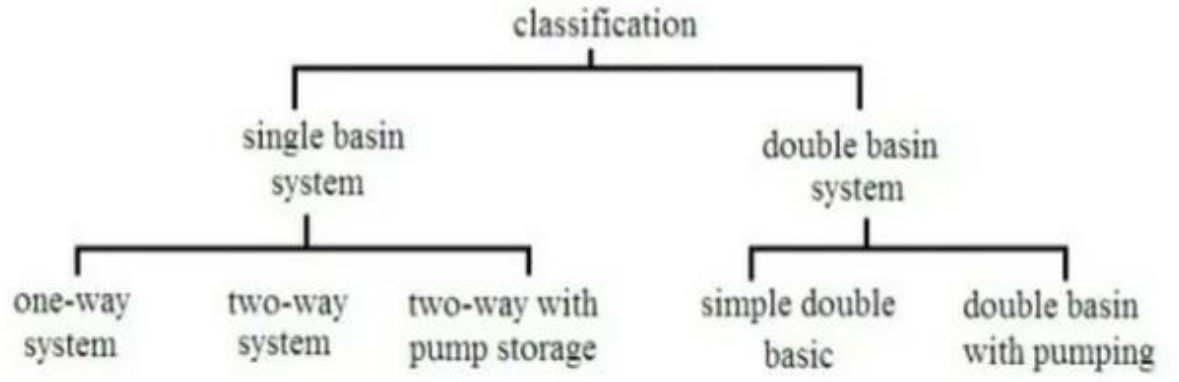
- Free from pollution
- Inexhaustible
- The construction does not uproot the villages or disturb the ecology of the place.
- No submerging of land occurs while constructing basins.
- Its operation is not adversely affected by variations of weather. Tide cycle is always definite.
- Basins can also be used for fish farming.

Limitations

- It can be harnessed only if natural sites are available.
- The suitable sites are bays, which are far from load centers. So transmission cost of power is high.
- Its operation is uneven as turbine have to work with varying water heads. The power output is variable.
- The construction period of a dam for basin is long. The basin and plant construction is costly.
- Sedimentation and silt deposition in the basin take place, which is costly to be removed.
- Power output varies with Lunar cycle.
- Life of equipment affected by the corrosive nature of sea water.

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Classification of tidal Power Plants



Single Basin System

- ***Ebb generation***:

Sluice way is opened to fill the basin during high tides. When water level in the basin reaches the highest level of the tide, sluice gate is closed. The turbine is started only when the water in the sea is at falling tide level. Electric power generation continues until the head is reduced to the minimum turbine operating level. The power output from this system is intermittent as power can be generated only during low tides.

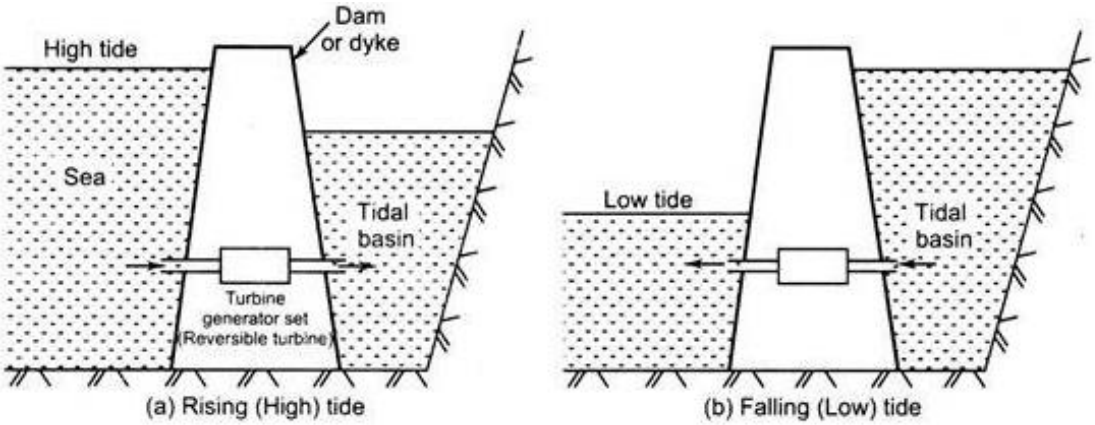
- ***Single Tide Cycle (flood tide)***

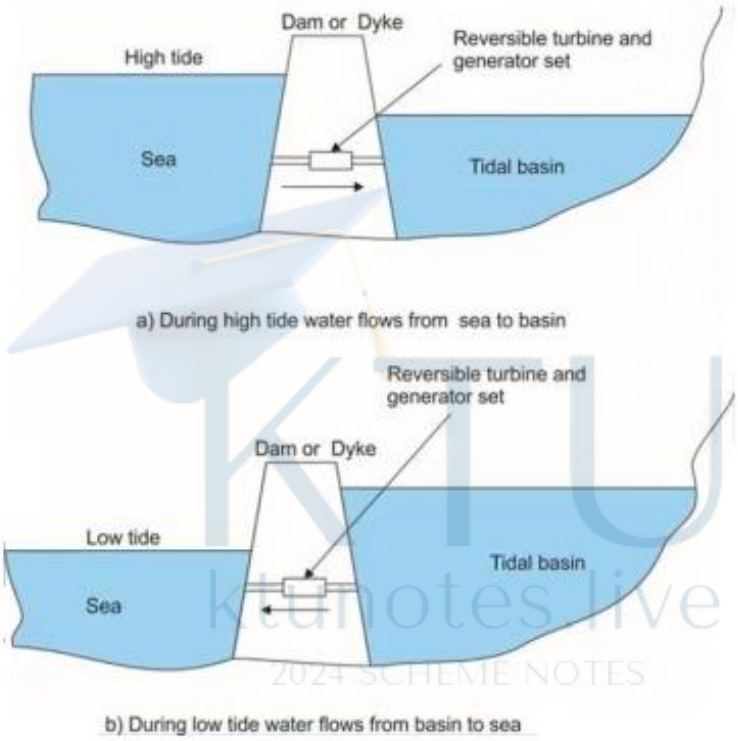
The basin is filled through the turbine which generate at flood tide. The water is made to enter into the basin through turbines during high tides. When high tide period is over, the sea level starts falling, which reduces the head on the turbines. The power generation is stopped as sea level reduces with the onset of low tides. The basin is drained of water through the sluiceways as sea level is lower during low tides. So power output is intermittent.

- ***Single Basin - Double Cycle System***

Here reversible turbines are installed. It is a combination of ebb tide and flood tide. Power is generated both during emptying and filling of the basin. Filling of basin is done during high tides when water level in basin is low. The water is made to flow through the turbines into basin to run generators. Emptying of the basin is done during the low tides when sea level is low and level in the basin is high. The water is made to flow out of the basin through turbines into sea to run generators. The flow is therefore used in both directions to drive the reversible turbines. Continuous power generation is not possible as it takes some time to change from filling to emptying mode of basin.

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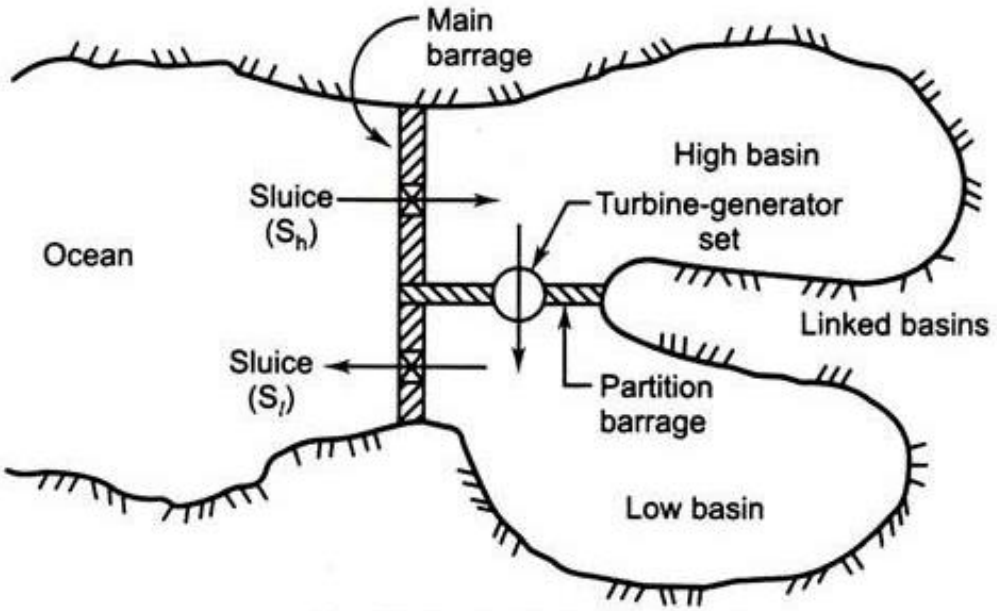
Double-Basin System:

- ***Double Basin – Linked Basin Arrangement***

A large basin is converted into two basins of suitable dimensions, one at higher level is called high basin and the other low basin. the scheme consists of three barrages', one separating high basin from the sea containing the filling gates (are opened when the sea level is higher than the upper basin.), another separating low basin from the sea and containing the emptying gates (opened only when the sea level is lower than the lower basin.).

The third barrage separates high basin from low basin and contains the power house. The head on the turbine is the difference in elevation between upper and lower basin.

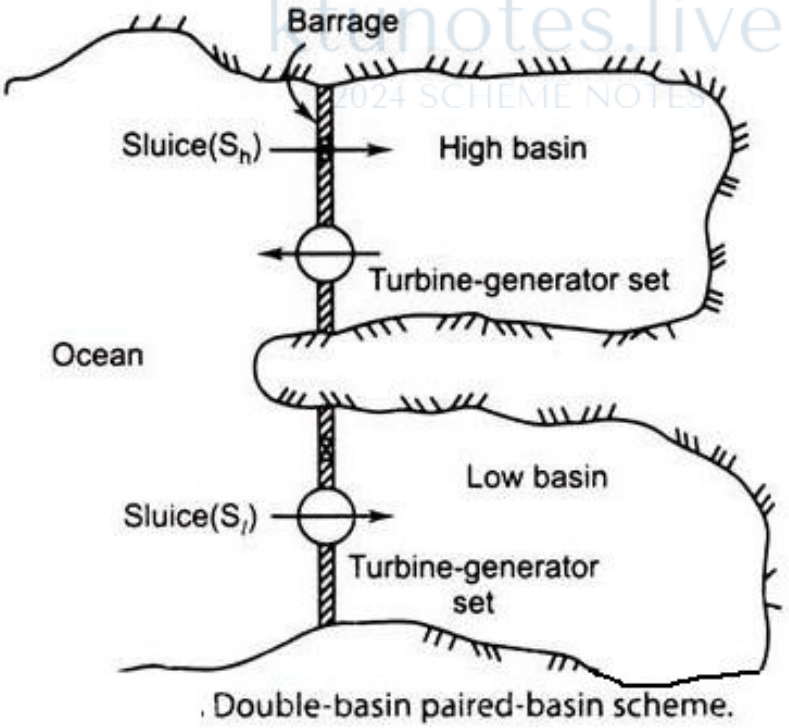
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Double-basin, linked-basin scheme.

- ***Double Basin – Paired Basin Arrangement***

Two single basin single effect separate schemes located at a distance from each other. The locations are selected that there is difference in tidal phase between them. They never exchange water, but are interconnected electrically. One basin generates electrical energy during the filling process while the other during the emptying process.



. Double-basin paired-basin scheme.

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Advantages of Tidal Power

- The biggest advantage of the tidal power is inexhaustible.
- It is completely independent of the rain.
- Tidal power generation is free from pollution, as it does not use any fuel and also does not produce any unhealthy waste like gases, ash or atomic residue.
- These power plants do not demand a large area of valuable land
- The net-cost of power generated is quite low

Disadvantages of Tidal Power

- The variability in output caused by the variations in the tidal range.
- The tidal range is highly variable and thus the turbine has to work on a wide range of head variations. This affects the efficiency of the plant.
- Suitable only at sites having a tidal range of 5 m or more
- Requirement of large volume of water flow at low head needs parallel operation of many turbines
- Sea water is corrosive.
- Construction in sea or near the sea is difficult.
- Cost is not favourable compared to the other sources of energy.
- The power transmission cost is high because the tidal power plants are located away from load centres.
- Occurrence of tides may not coincide with times of peak demand

Previous University Questions

1. List the factors that affect biogas generation.
2. With neat diagram, explain how power can be extracted using satellite stations.
3. With a neat diagram, explain the working of Janata model/Fixed drum biogas plant
4. Discuss the methods of urban waste to energy conversion.
5. Explain the factors affecting biogas generation
6. Explain the process of Pyrolysis in biomass conversion.
7. Draw the layout of a biogas plant and explain the digestion process.
8. Explain the concept of urban waste to energy conversion.
9. Explain the term “tidal range” with the help of a figure.
10. List three advantages and disadvantages of tidal power plants.
11. Classify tidal power plants based on the type of basin used and explain their working with relevant figures.
12. List the advantages and limitations of tidal power plants.
13. Describe the principle and various components of tidal power plants.
14. List three advantages of tidal power plant.
15. With neat diagrams, explain single basin double effect tidal power generation scheme for one tidal cycle.
16. With neat diagrams, explain single basin single effect tidal power generation scheme for one tidal cycle
17. Explain the principle of Tidal energy harnessing
18. With relevant figures, explain the single basin and double basin tidal power plants.

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19. Discuss the advantages and limitations of tidal power plants.
20. Enumerate the key criteria for selecting suitable sites for a tidal plant.
21. Explain the double basin schemes for the tidal energy conversion. Highlight their advantage over single basin schemes.
22. Classify single-basin tidal power generation systems based on their mode of operation. Describe each mode and its relevance in terms of power generation.
23. A double basin arrangement can increase the amount of energy harvested from a tidal power generation system. Justify this statement.
24. Differentiate the working principle of a tidal energy conversion system and an ocean thermal energy conversion system. State the advantages and disadvantages of these schemes.
25. How is a single-basin tidal power generation classified based on the mode of power generation?
26. Classify tidal power plants based on the type of basin used with relevant figures.

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